

Multi-Layered SOC Lab: Threat Detection & Automated Malware Remediation

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May 2026

Tech Stack: Wazuh (SIEM), Suricata (IDS), VirusTotal API, Ubuntu (Endpoint), Kali Linux (Manager), Diamorphine.

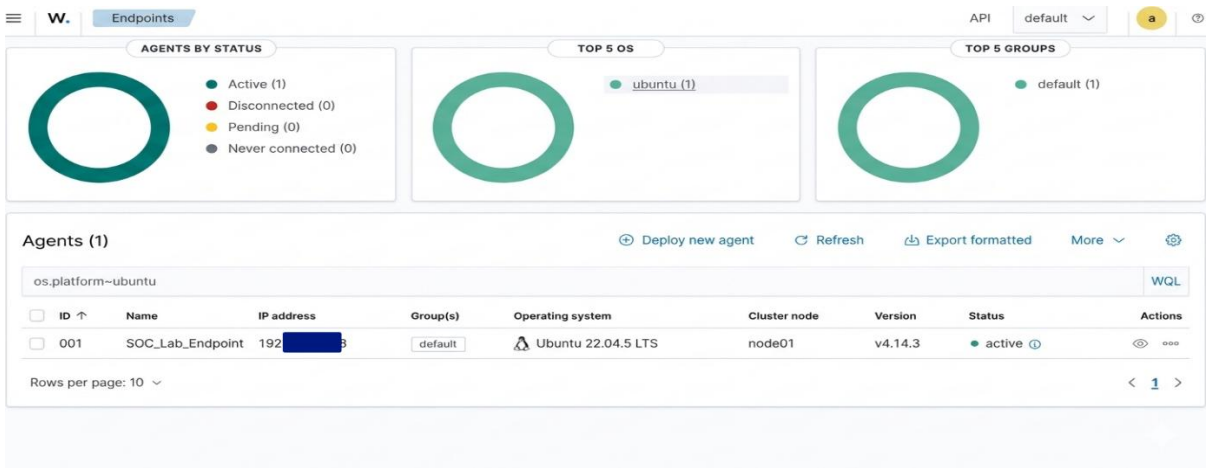
1. Executive Summary

This project demonstrates the deployment of a functional Security Operations Center (SOC) designed to detect host-based, network-based, identity-based and file-based threats. By integrating Wazuh, Suricata, and VirusTotal, I created a unified visibility pipeline that identifies kernel-level obfuscation (rootkits), malicious network patterns, unauthorized access attempts, and automated malware remediation within a virtualized lab environment.

2. Infrastructure Architecture

The lab environment was built using a private virtual network to simulate an enterprise internal LAN.

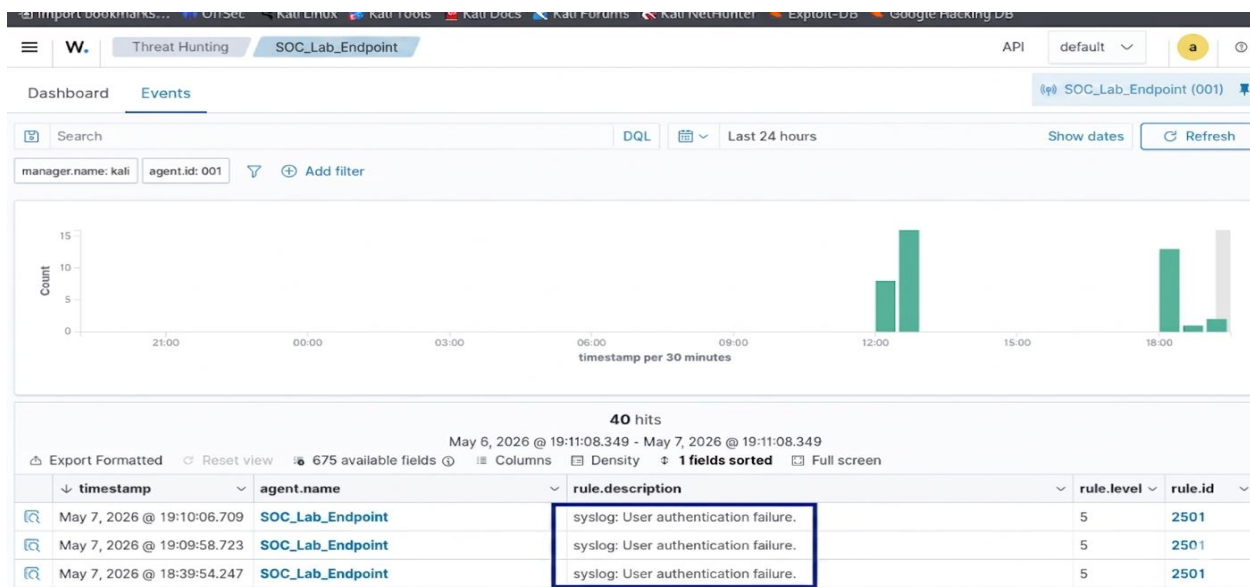
- **Wazuh Manager:** Kali Linux (Centralized Intelligence).
- **Managed Endpoint: Agent Name:** SOC_Lab_Endpoint (Ubuntu 22.04 LTS)
- **Connectivity:** The SOC_Lab_Endpoint agent communicates with the Manager over a dedicated virtual bridge. Connectivity was verified via the Wazuh Dashboard, showing the agent status as "Active."



3. Identity & Access Monitoring: Syslog Analysis

I established baseline visibility for system access.

- **Attack Simulation:** Performed multiple failed login attempts on the SOC_Lab_Endpoint agent.
- **Detection Result:** The Wazuh Dashboard successfully generated "User authentication failure" alerts Rule level 3 and Rule.id 2501. This provided real-time visibility into potential brute-force attacks.



4. Host-Based Security: Rootkit Emulation & Detection

The objective was to test the SIEM's ability to see "invisible" threats using the Diamorphine rootkit.

- **Attack Phase:** Executed a stealth attack to hide the rsyslogd process. Got a PID using **ps aux | head** and run the kill signal Using **kill -31 PID**, the process remained active in memory but became invisible to **ps aux**
- **Defense Phase:** Hardened the SOC_Lab_Endpoint agent by changing the rootcheck frequency to 30 seconds for rapid integrity audits.
- **Detection Result:** Wazuh Manager identified the discrepancy between the kernel's process table and system calls, triggering a Level 11 High Severity Alert (Rule 521) for a hidden process.

```

root@osboxes:/home/osboxes/Diamorphine# ps aux | head
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.0  0.3 168036 13456 ?        Ss   13:19   0:06 /sbin/init splash
root         2  0.0  0.0      0     0 ?        S    13:19   0:00 [kthreadd]
root         3  0.0  0.0      0     0 ?        I<   13:19   0:00 [rcu_gp]
root         4  0.0  0.0      0     0 ?        I<   13:19   0:00 [rcu_par_gp]
root         6  0.0  0.0      0     0 ?        I<   13:19   0:00 [kworker/0:0H-kblockd]
root         9  0.0  0.0      0     0 ?        I<   13:19   0:00 [mm_percpu_wq]
root        10  0.0  0.0      0     0 ?        S    13:19   0:00 [rcu_tasks_rude_]
root        11  0.0  0.0      0     0 ?        S    13:19   0:00 [rcu_tasks_trace]
root        12  2.3  0.0      0     0 ?        S    13:19   3:13 [ksoftirqd/0]
root@osboxes:/home/osboxes/Diamorphine# lsmod | grep diamorphine
kill -63 1
lsmod | grep diamorphine
diamorphine 16384 0
root@osboxes:/home/osboxes/Diamorphine# ps auxw | grep rsyslogd | grep -v grep
syslog      649  0.0  0.1 222404 4964 ?        Ssl  13:20   0:00 /usr/sbin/rsyslogd -n -iNONE
root@osboxes:/home/osboxes/Diamorphine# kill -31 649
root@osboxes:/home/osboxes/Diamorphine# ps auxw | grep rsyslog | grep -v grep
root@osboxes:/home/osboxes/Diamorphine#

```

The screenshot shows the Wazuh Threat Hunting interface. It displays a list of 640 hits for rule 521, which is 'Possible kernel level rootkit'. The hits are from the agent 'SOC_Lab_Endpoint' and occurred on May 7, 2026, at 20:45:30.773, 20:45:30.700, 20:45:30.621, and 20:45:29.073. The rule level is 11 and the rule ID is 521.

timestamp	agent.name	rule.description	rule.level	rule.id
May 7, 2026 @ 20:45:30.773	SOC_Lab_Endpoint	Possible kernel level rootkit	11	521
May 7, 2026 @ 20:45:30.700	SOC_Lab_Endpoint	Possible kernel level rootkit	11	521
May 7, 2026 @ 20:45:30.621	SOC_Lab_Endpoint	Possible kernel level rootkit	11	521
May 7, 2026 @ 20:45:29.073	SOC_Lab_Endpoint	Possible kernel level rootkit	11	521

5. Network Security: Suricata IDS Integration

- **Implementation:** Installed Suricata with the Emerging Threats (ET) Open rule set.
- **The Pipeline:** Integrated Suricata's eve.json output into the SOC_Lab_Endpoint Agent's ingestion engine, merging network telemetry with host logs.
- **Validation:** Generated ICMP traffic via the Kali Linux terminal. The IDS successfully flagged the reconnaissance attempt, providing a unified view on the Wazuh Dashboard.

```
osboxes@osboxes: /tmp
GNU nano 6.2 /var/ossec/etc/ossec.conf
<active-response>
  <disabled>no</disabled>
  <ca_store>etc/wpk_root.pem</ca_store>
  <ca_verification>yes</ca_verification>
</active-response>

<!-- Choose between "plain", "json", or "plain,json" for the format of intern
<logging>
  <log_format>plain</log_format>
</logging>

</ossec_conf>

<ossec_config>
  <localfile>
    <log_format>journald</log_format>
    <location>journald</location>
  </localfile>

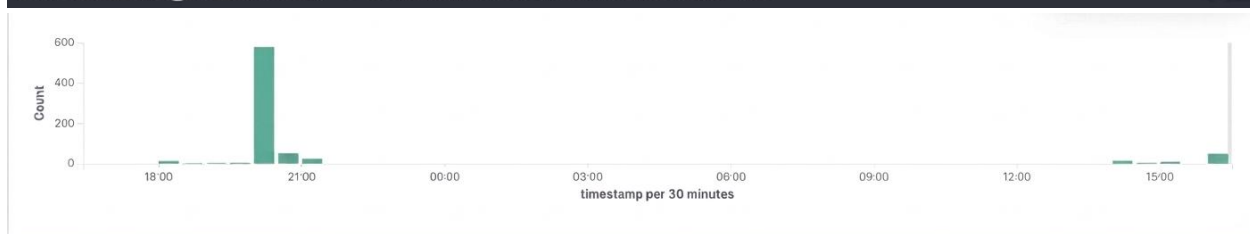
  <localfile>
    <log_format>syslog</log_format>
    <location>/var/ossec/logs/active-responses.log</location>
  </localfile>

  <localfile>
    <log_format>syslog</log_format>
    <location>/var/log/dpkg.log</location>
  </localfile>

  <localfile>
    <log_format>json</log_format>
    <location>/var/log/suricata/eve.json</location>
  </localfile>
</ossec_config>
```

```
Session Actions Edit View Help
kali)-[~]
$ ping -c 20 192.168.1.8
PING 192.168.1.8 (192.168.1.8) 56(84) bytes of data.
64 bytes from 192.168.1.8: icmp_seq=1 ttl=64 time=1.06 ms
64 bytes from 192.168.1.8: icmp_seq=2 ttl=64 time=0.898 ms
64 bytes from 192.168.1.8: icmp_seq=3 ttl=64 time=1.43 ms
64 bytes from 192.168.1.8: icmp_seq=4 ttl=64 time=1.42 ms
64 bytes from 192.168.1.8: icmp_seq=5 ttl=64 time=1.10 ms
64 bytes from 192.168.1.8: icmp_seq=6 ttl=64 time=1.35 ms
64 bytes from 192.168.1.8: icmp_seq=7 ttl=64 time=0.977 ms
64 bytes from 192.168.1.8: icmp_seq=8 ttl=64 time=1.77 ms
64 bytes from 192.168.1.8: icmp_seq=9 ttl=64 time=1.37 ms
64 bytes from 192.168.1.8: icmp_seq=10 ttl=64 time=0.923 ms
64 bytes from 192.168.1.8: icmp_seq=11 ttl=64 time=1.24 ms
64 bytes from 192.168.1.8: icmp_seq=12 ttl=64 time=1.03 ms
64 bytes from 192.168.1.8: icmp_seq=13 ttl=64 time=1.31 ms
64 bytes from 192.168.1.8: icmp_seq=14 ttl=64 time=1.42 ms
64 bytes from 192.168.1.8: icmp_seq=15 ttl=64 time=1.46 ms
64 bytes from 192.168.1.8: icmp_seq=16 ttl=64 time=1.18 ms
64 bytes from 192.168.1.8: icmp_seq=17 ttl=64 time=1.36 ms
64 bytes from 192.168.1.8: icmp_seq=18 ttl=64 time=1.24 ms
64 bytes from 192.168.1.8: icmp_seq=19 ttl=64 time=1.17 ms
64 bytes from 192.168.1.8: icmp_seq=20 ttl=64 time=0.955 ms

— 192.168.1.8 ping statistics —
20 packets transmitted, 20 received, 0% packet loss, time 19226ms
rtt min/avg/max/mdev = 0.898/1.232/1.771/0.218 ms
```



750 hits

May 7, 2026 @ 16:24:48.281 - May 8, 2026 @ 16:24:48.281

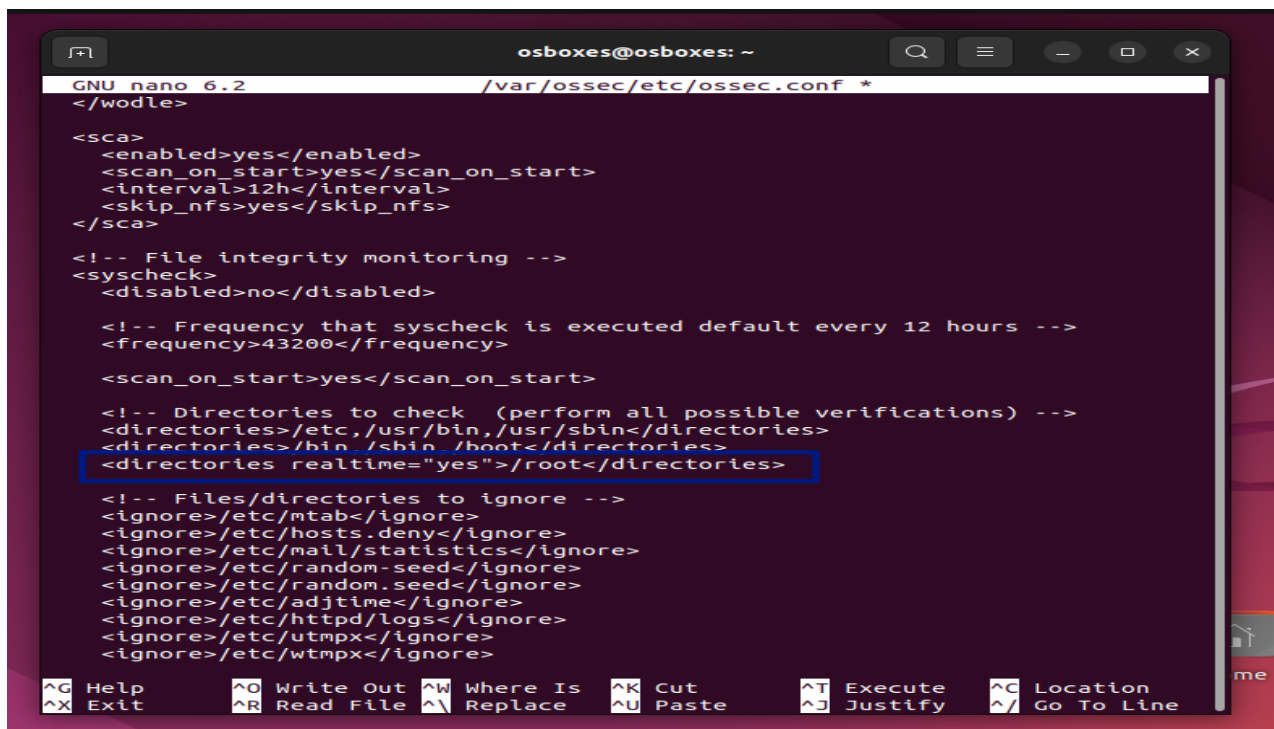
Export Formatted Reset view 684 available fields Columns Density 1 fields sorted Full screen

timestamp	agent.name	rule.description	rule.level	rule.id
May 8, 2026 @ 16:22:08.468	SOC_Lab_Endpoint	Suricata: Alert - GPL ICMP_INFO PING +NIX	3	86601
May 8, 2026 @ 16:22:08.466	SOC_Lab_Endpoint	Suricata: Alert - GPL ICMP_INFO PING +NIX	3	86601
May 8, 2026 @ 16:22:06.465	SOC_Lab_Endpoint	Suricata: Alert - GPL ICMP_INFO PING +NIX	3	86601
May 8, 2026 @ 16:22:06.462	SOC_Lab_Endpoint	Suricata: Alert - GPL ICMP_INFO PING +NIX	3	86601

6. Automated Response: VirusTotal & Malware Remediation

Implemented an **Active Response** loop to handle malicious files automatically.

- **File Integrity Monitoring (FIM):** Configured SOC_Lab_Endpoint agent to monitor the /root directory for new files.
- **API Integration:** Connected Wazuh Manager to the VirusTotal API to analyze file hashes detected on the SOC_Lab_Endpoint agent via FIM.
- **Threat Trigger:** Created a custom rule (**87105**) that fires only when VirusTotal returns a positive malware result.
- **Validation:** I downloaded the **EICAR** test file to the SOC_Lab_Endpoint /root directory. Within seconds, the file was identified, verified as malicious, and deleted.



```
osboxes@osboxes: ~  
GNU nano 6.2 /var/ossec/etc/ossec.conf *  
</wodle>  
  
<sca>  
  <enabled>yes</enabled>  
  <scan_on_start>yes</scan_on_start>  
  <interval>12h</interval>  
  <skip_nfs>yes</skip_nfs>  
</sca>  
  
<!-- File integrity monitoring -->  
<syscheck>  
  <disabled>no</disabled>  
  
  <!-- Frequency that syscheck is executed default every 12 hours -->  
  <frequency>43200</frequency>  
  
  <scan_on_start>yes</scan_on_start>  
  
  <!-- Directories to check (perform all possible verifications) -->  
  <directories>/etc,/usr/bin,/usr/sbin</directories>  
  <directories>/bin /sbin /boot</directories>  
  <directories realtime="yes">/root</directories>  
  
  <!-- Files/directories to ignore -->  
  <ignore>/etc/mtab</ignore>  
  <ignore>/etc/hosts.deny</ignore>  
  <ignore>/etc/mail/statistics</ignore>  
  <ignore>/etc/random-seed</ignore>  
  <ignore>/etc/random.seed</ignore>  
  <ignore>/etc/adjtime</ignore>  
  <ignore>/etc/httpd/logs</ignore>  
  <ignore>/etc/utmpx</ignore>  
  <ignore>/etc/wtmpx</ignore>  
</syscheck>  
</ossec>
```



```

GNU nano 8.7.1 /var/ossec/etc/ossec.conf *
<skip_nfs>yes</skip_nfs>

<ignore>/var/lib/containerd</ignore>
<ignore>/var/lib/docker/overlay2</ignore>
</rootcheck>

<wodle name="cis-cat">
  <disabled>yes</disabled>
  <timeout>1800</timeout>
  <interval>1d</interval>
  <scan-on-start>yes</scan-on-start>

  <java_path>wodles/java</java_path>
  <ciscat_path>wodles/ciscat</ciscat_path>
</wodle>

<!-- Osquery integration -->
<wodle name="osquery">
  <disabled>yes</disabled>
  <run_daemon>yes</run_daemon>
  <log_path>/var/log/osquery/osqueryd.results.log</log_path>
  <config_path>/etc/osquery/osquery.conf</config_path>
  <add_labels>yes</add_labels>
</wodle>

<integration>
  <name>virustotal</name>
  <api_key>09b42c5b5afc29c35ea68005[REDACTED]</api_key>
  <rule_id>100200,100201</rule_id>
  <alert_format>json</alert_format>
</integration>

<!-- System inventory -->
<wodle name="syscollector">

```

```
Session Actions Edit View Help
GNU nano 8.7.1 /var/ossec/etc/ossec.conf *
<executable>netsh.exe</executable>
<timeout_allowed>yes</timeout_allowed>
</command>

<command>
<name>remove-threat</name>
<executable>remove-threat.sh</executable>
</command>

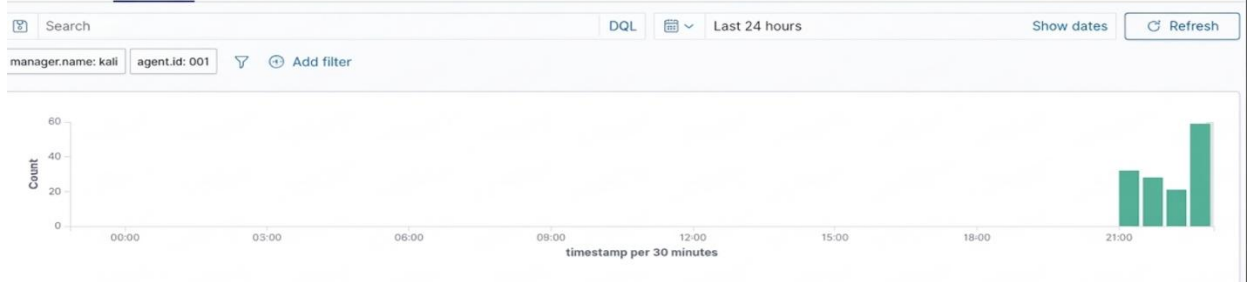
<!--
<active-response> Disconnected (0) Critical severity High severity
active-response options here
</active-response>
-->

Rule level 15 or higher Rule level 12 to 14

<active-response>
<command>remove-threat</command>
<location>local</location>
<rules_id>87105</rules_id>
</active-response>

Log analysis -->
<logfile>
<log_format>command</log_format>
<command>df -P</command>
<frequency>360</frequency>
</logfile>

Malware Detection Threat Hunting
Check indicators of compromise triggered by malware infections or Browse through security alerts issues and their status
```



140 hits

May 9, 2026 @ 22:50:13.058 - May 10, 2026 @ 22:50:13.058

Export Formatted

Reset view

694 available fields

Columns

Density

1 fields sorted

Full screen

timestamp	agent.name	rule.description	rule.level	rule.id
May 10, 2026 @ 22:49:47.8...	SOC_Lab_Endpoint	File deleted.	7	553
May 10, 2026 @ 22:49:47.7...	SOC_Lab_Endpoint	VirusTotal: Alert - /root/eicar.com - 66 engines detected this file	12	87105
May 10, 2026 @ 22:49:38.9...	SOC_Lab_Endpoint	PAM: Login session closed.	3	5502
May 10, 2026 @ 22:49:37.3...	SOC_Lab_Endpoint	File added to /root	7	100201
May 10, 2026 @ 22:49:36.9...	SOC_Lab_Endpoint	PAM: Login session opened	3	5501

```
osboxes@osboxes: ~  
osboxes@osboxes:~$ sudo curl -Lo /root/eicar.com https://secure.eicar.org/eicar.com  
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current  
             Dload  Upload   Total     Spent    Left     Speed  
100    68  100    68    0     0    71      0  --:--:-- --:--:-- --:--:--    71  
osboxes@osboxes:~$ sudo ls -la /root/eicar.com  
ls: cannot access '/root/eicar.com': No such file or directory  
osboxes@osboxes:~$
```

7. Conclusion

This lab validates a defense-in-depth strategy. While the rootkit (Diamorphine) successfully fooled local OS utilities and network probes (ICMP) tested the perimeter, the SOC's ability to cross-reference kernel data, inspect network packets, and autonomously remediate malware through API integrations ensures that even sophisticated attacks are identified and neutralized.